

AP CALCULUS AB - UNIT 1

1.8 Determining Limits Using the Squeeze Theorem

Lesson Overview

- State all hypotheses of the Squeeze Theorem.
- Create useful upper and lower bounds from absolute values and trigonometric inequalities.
- Distinguish persistent oscillation from shrinking oscillation.
- Write a complete AP-style squeeze justification.

Key Knowledge and Vocabulary

Ordering. Find functions $g(x) \leq f(x) \leq h(x)$ on a deleted neighborhood of the target.

Shared outer limit. If $\lim g = \lim h = L$, then $\lim f = L$.

Absolute-value route. If $|f(x) - L| \leq B(x)$ and $B(x) \rightarrow 0$, then $f(x) \rightarrow L$.

Bounded times vanishing. If $|u(x)| \leq M$ and $v(x) \rightarrow 0$, then $u(x)v(x) \rightarrow 0$.

One-sided care. Inequalities must hold on the side being considered; multiplying by negative quantities reverses order.

Explanation and Strategy

Step 1. Start from a known bound such as $-1 \leq \sin u \leq 1$ or $|\cos u| \leq 1$.

Step 2. Multiply by a nonnegative shrinking factor or move to an absolute-value inequality.

Step 3. End by naming the theorem and the common outer limit; merely saying “squeezed” is incomplete.

Solved Examples

Worked Example: Shrinking oscillation

Evaluate $\lim_{x \rightarrow 0} x^2 \sin(1/x)$.

Solution. Because $|\sin(1/x)| \leq 1$, $|x^2 \sin(1/x)| \leq x^2 \rightarrow 0$. Therefore the limit is 0 by the Squeeze Theorem.

Worked Example: Shifted center

Evaluate $\lim_{x \rightarrow 3} (x - 3) \cos\left(\frac{1}{x-3}\right)$.

Solution. The absolute value is at most $|x - 3|$, which tends to 0. Thus the limit is 0.

Worked Example: Sine anchor from geometry

Given $\cos x \leq \frac{\sin x}{x} \leq 1$ for small positive x , find the right-hand limit at 0.

Solution. Both outer functions approach 1, so $\lim_{x \rightarrow 0^+} \sin x/x = 1$. Symmetry extends this to the two-sided limit.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Given $-x^2 \leq f(x) \leq x^2$ near 0, find $\lim_{x \rightarrow 0} f(x)$.

2. Evaluate $\lim_{x \rightarrow 0} x \sin(1/x)$.

3. Evaluate $\lim_{x \rightarrow 2} (x - 2)^2 \cos(1/(x - 2))$.

4. What two facts must accompany the inequality $g \leq f \leq h$ to use the theorem?

Medium Level

Apply the idea in one or two connected steps.

5. Evaluate $\lim_{x \rightarrow 0} |x|^{3/2} \sin(7/x)$.

6. If $|f(x) - 4| \leq 3|x - 1|$ near 1, find $\lim_{x \rightarrow 1} f(x)$.

7. Evaluate $\lim_{x \rightarrow 0} x^2 \cos(e^{1/x^2})$.

8. Explain why the bound $-1 \leq \sin(1/x) \leq 1$ does not determine $\lim_{x \rightarrow 0} \sin(1/x)$.

Hard Level

Connect representations, conditions, and justification.

9. Evaluate $\lim_{x \rightarrow 0} \frac{x^3}{|x|} \sin(1/x)$.

10. If $0 \leq f(x) \leq \frac{(x-5)^2}{1+(x-5)^2}$, find the limit at 5.

11. Evaluate $\lim_{x \rightarrow 0} x \arctan(1/x)$.

12. Prove $\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x} = 0$ without using derivatives.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Suppose $|f(x)| \leq |x|$ and $|g(x)| \leq x^2$ near 0. Find $\lim_{x \rightarrow 0} \frac{f(x)g(x)}{x}$ if defined for $x \neq 0$.

14. Evaluate $\lim_{x \rightarrow 0} x^2 \left\lfloor \frac{1}{|x|} \right\rfloor$.

15. If $|f(x) - L| \leq |x - c| |\ln |x - c||$ near c , determine the limit of f at c .

Challenge Level

Synthesize, construct, generalize, or prove.

16. Prove: if u is bounded near c and $v(x) \rightarrow 0$, then $u(x)v(x) \rightarrow 0$.

17. Find $\lim_{x \rightarrow 0} x^2 \sin(1/x) \cos(e^{1/x^4})$ and justify every bound.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. If $-2|x - 1| \leq f(x) \leq 5|x - 1|$ near 1, then $\lim_{x \rightarrow 1} f(x)$ is

- (A) -2 (B) 0
(C) 5 (D) not determined

2. Which bound is useful for $x^3 \cos(1/x)$ near 0?

- (A) $|x^3 \cos(1/x)| \leq |x|^3$ (B) $|x^3 \cos(1/x)| \geq 1$
(C) $\cos(1/x) \rightarrow 0$ (D) $x^3 \geq 1$

3. Why can the theorem not be applied to $-1 \leq f(x) \leq 1$ alone?

- (A) The inequality is false. (B) The outer limits are not equal.
(C) f must be polynomial. (D) The target must be 1.

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A graphing calculator shows violent oscillation for $x \sin(1/x)$ near 0, but the envelope narrows to the axis.

The limit is

- (A) -1 (B) 0
(C) 1 (D) DNE

5. Numerically, $x^2 \sin(10^6/x)$ is difficult to sample near 0. Which theoretical statement resolves the limit?

- (A) $|x^2 \sin(10^6/x)| \leq x^2$ (B) $\sin(10^6/x) \rightarrow 1$
(C) $x^2 \rightarrow 1$ (D) The frequency is finite

Free-Response Practice – No Calculator

Let $F(x) = (x - 2)^2 \sin\left(\frac{3}{x-2}\right)$ for $x \neq 2$.

(a) Find an upper bound for $|F(x)|$. [1 points]

(b) Use the Squeeze Theorem to find $\lim_{x \rightarrow 2} F(x)$. [3 points]

(c) What value should be assigned to $F(2)$ to make the extended function continuous? [1 points]

Free-Response Practice – Calculator Required

A calculator graph of $G(x) = x \sin(1/x^2)$ near 0 appears increasingly dense.

(a) Use the graph to conjecture the limit. [1 points]

(b) Give a non-calculator proof. [3 points]

(c) Explain why calculator sampling alone may miss the full oscillation.

[2 points]